



# Copula-based dynamic networks for forecasting stock market volatility

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## ABSTRACT

This study enhances volatility forecasting by integrating copula-based methods with a dynamic network log-ARCH model. The D-vine copula quantile regression and copula entropy methods, which capture nonlinear dependencies and select key predictors, are used to construct a high-dimensional financial network. The model incorporates time-lagged volatility and information from neighboring stocks, enabling efficient risk propagation. Applied to Dow Jones Index constituents (2012–2024), our approach improves forecasting accuracy while reducing reliance on hyperparameters. We also find that the low-volatility premium coincides with lower downside risk, with low- and high-volatility stocks exhibiting equal or lower return standard deviations across all networks. Notably, the copula-based network identifies stocks that enhance alpha while mitigating portfolio risk, offering a robust tool for portfolio construction and risk management.

## 1. Introduction

Stock volatility is a fundamental indicator of market fluctuation and risk, informing investment decisions and risk management. Among various modeling techniques, the generalized autoregressive conditional heteroscedasticity (GARCH) model and its extensions were widely used in the literature (Bollerslev, 1986; Andersen and Bollerslev, 1998; Francq and Zakoian, 2019; Tian and Ji, 2022). However, due to persistent volatility, large shocks, and skewed distributions in financial returns, the log-ARCH model (Geweke, 1986; Pourkhanali et al., 2023), instead, offers better flexibility and precision in modeling volatility dynamics. Recent studies also integrate spatial econometric approaches into volatility modeling (e.g., Sato and Matsuda, 2017; Otto and Schmid, 2018) that relies on a spatial proximity matrix. We propose a novel method for defining proximity using copula-based approaches.

Several network-based models (e.g., k-NN), assume a stock's volatility depends on both its own history and the volatilities of connected stocks (Otto and Schmid, 2019; Zareei, 2019; Liu et al., 2021; Tafakori et al., 2022; Rašiová and Arendás, 2023). These networks are defined using similarity measures such as correlation, Euclidean distance, or time series distances extracted from autoregressive (AR) models (Mattera and Otto, 2024). While spatiotemporal models incorporate inter-stock effects, defining influential neighbors is still challenging. Prior studies use metrics based on financial indicators (Asgharian et al., 2013), balance sheet distances (Fernández Avilés et al., 2012), or production linkages (Ando et al., 2023), but often impose strong assumptions and miss full dependency structures, leading to potential model misspecification.

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Copula models offer a flexible alternative by separating marginal behavior from dependence structures, effectively capturing nonlinear and asymmetric relationships. They have been used for dependence dynamics (Chen and Fan, 2004; Pourkhanali et al., 2016), modeling exchange rate co-movements (Patton, 2002), and constructing portfolios (Xu, 2005). More recent advances include copula-based clustering (Oh and Patton, 2023), flexible copula estimation (Hu, 2006; Nagler et al., 2019), and vine copulas for portfolio risk (Sommer et al., 2023).

Building on prior work, we adopt a D-vine copula framework to model volatility. As a subclass of regular vine copulas, D-vines decompose multivariate dependencies into conditional bivariate copulas, allowing for flexible, asymmetric, and tail-dependent modeling without restrictive distributional assumptions (Kraus and Czado, 2017). This is well-suited to capturing complex, dynamic relationships in financial networks.

To further account for short-term nonlinear dependencies, we incorporate copula entropy, an information-theoretic measure rooted in Shannon (1948). Unlike traditional metrics like correlation or variance, copula entropy quantifies uncertainty in dependence structures and enhances risk assessment (Saleh, 2007; Wu et al., 2009). It has been applied in finance for credit modeling and portfolio optimization (Ane and Kharoubi, 2003; Dempster et al., 2007), and also aids in ranking copula families.

We contribute to existing literature and address key gaps in three ways.

First, our model captures nonlinear and asymmetric dependencies. This study employs copula-based methods to capture complex distributional relationships, allowing for flexible, nonlinear, and asymmetric dependence structures among stocks. Unlike traditional approaches, such as Mattera and Otto (2024), which rely on linear correlation or distance-based proximity matrices (e.g., Euclidean distance or fixed k-nearest neighbors), the copula-based framework can more effectively model tail dependence and non-Gaussian features in return distributions. This offers a more realistic representation of market co-movements, especially during periods of stress (Patton, 2006; Jondeau and Rockinger, 2006).

Second, we model long and short-term market dynamics with time-varying networks. By integrating long-term dependence via D-vine copulas with short-term dynamics identified through copula entropy, we construct a time-varying financial network that adapts to evolving market conditions. In contrast, Mattera and Otto (2024) focus on fixed network structures and combined forecasts. They use average minimum-variance weighting, or constrained OLS (Timmermann, 2006), without considering structural breaks or changes in market regimes. These static frameworks often overlook the dynamically updating relationships between assets, which is critical in financial market forecasting (Engle, 2002; Barigozzi et al., 2021). Our approach addresses this gap by allowing the network structure to evolve over time, enhancing forecasting accuracy and robustness.

Third, this study provides practical implications for asset selection and confirms the low-volatility anomaly. The proposed framework doubles as a practical asset selection tool by identifying stocks that exhibit low predicted volatility and high returns, offering insights into the low-volatility anomaly documented in asset pricing literature (Baker et al., 2011; Frazzini and Pedersen, 2014). Unlike studies that categorize stocks based on arbitrary groupings or backward-looking volatility measures (Ang et al., 2006), our approach forms groups based on forward-looking volatility forecasts derived from the copula-based network model. This method improves the identification of risk-adjusted investment opportunities and supports dynamic portfolio construction strategies.

We use the Dow Jones Industrial Average (DJIA) Index constituents as the research field following Mattera and Otto (2024). We find that our proposed model outperforms benchmark models in forecasting accuracy. Importantly, the copula models' performance remains stable even under turbulent conditions during the Global Financial Crisis (GFC), underscoring its practical utility. Portfolio analysis further confirms the volatility anomaly and the practical implication of our model, that 0.014% – 0.018% daily abnormal returns are generated by the long-short strategy.

The remainder of the paper is structured as follows. Section 2 puts forward hypothesis development, Section 3 outlines the sample, spatiotemporal log-ARCH model, D-vine copula, and copula entropy methods. Section 4 presents empirical results and benchmarking analysis. Section 5 concludes with key insights and implications.

## 2. Hypotheses development

By integrating copula methods into a network log-ARCH framework, we expect improved accuracy in capturing volatility spillovers and forecasting future volatility. These methodological enhancements should lead to superior performance compared to conventional models, especially under volatile or turbulent market conditions.

Based on the above, we develop the following hypothesis:

**H1.** The copula-based network models outperform log-ARCH models and k-NN models in forecasting stock volatility.

While D-vine copula model is effective in capturing long-term, stable, and hierarchical dependency structures, they are computationally intensive and less responsive to recent market changes. In contrast, copula entropy offers a computationally efficient, data-driven mechanism to detect short-term, nonlinear dependencies and rapidly evolving patterns in asset relationships. By combining D-vine copula and copula entropy, the hybrid model leverages the long-term insights from D-vine copula and the short-term adaptability of copula entropy, which is particularly valuable in volatile or regime-switching markets.

Based on the above, we propose hypothesis 2:

**H2.** The hybrid model combining D-vine copula and copula entropy outperforms the other models.

**Table 1**

Descriptive statistics of daily stock log returns in Dow Jones Industrial index from October 2012 to June 2024.

| Company                  | Ticker | Mean    | St.D.  | Min     | Max    |
|--------------------------|--------|---------|--------|---------|--------|
| Apple                    | AAPL   | 0.0009  | 0.0177 | -0.1377 | 0.1132 |
| Amgen                    | AMGN   | 0.0006  | 0.0152 | -0.1008 | 0.1117 |
| Amazon                   | AMZN   | 0.0009  | 0.0206 | -0.1514 | 0.1462 |
| American Express         | AXP    | 0.0006  | 0.0180 | -0.1604 | 0.1979 |
| Boeing                   | BA     | 0.0004  | 0.0228 | -0.2724 | 0.2177 |
| Caterpillar              | CAT    | 0.0005  | 0.0181 | -0.1541 | 0.0983 |
| Salesforce               | CRM    | 0.0006  | 0.0227 | -0.2199 | 0.2315 |
| Cisco                    | CSCO   | 0.0003  | 0.0164 | -0.1769 | 0.1480 |
| Chevron                  | CVX    | 0.0003  | 0.0172 | -0.2501 | 0.2049 |
| Goldman Sachs            | GS     | 0.0004  | 0.0178 | -0.1359 | 0.1620 |
| Home Depot               | HD     | 0.0008  | 0.0147 | -0.2206 | 0.1288 |
| Honeywell                | HON    | 0.0006  | 0.0143 | -0.1288 | 0.1404 |
| IBM                      | IBM    | 0.0002  | 0.0142 | -0.1375 | 0.1071 |
| Intel                    | INTC   | 0.0003  | 0.0194 | -0.1990 | 0.1783 |
| Johnson & Johnson        | JNJ    | 0.0004  | 0.0107 | -0.1058 | 0.0769 |
| JPMorgan Chase           | JPM    | 0.0006  | 0.0173 | -0.1621 | 0.1656 |
| Coca-Cola                | KO     | 0.0003  | 0.0108 | -0.1017 | 0.0628 |
| McDonald's               | MCD    | 0.0005  | 0.0118 | -0.1729 | 0.1666 |
| 3M                       | MMM    | 0.0002  | 0.0143 | -0.1386 | 0.1187 |
| Merck                    | MRK    | 0.0005  | 0.0130 | -0.1038 | 0.0990 |
| Microsoft                | MSFT   | 0.0009  | 0.0163 | -0.1595 | 0.1329 |
| Nike                     | NKE    | 0.0004  | 0.0178 | -0.2229 | 0.1444 |
| Procter & Gamble         | PG     | 0.0004  | 0.0109 | -0.0914 | 0.1134 |
| Travelers                | TRV    | 0.0005  | 0.0142 | -0.2332 | 0.1248 |
| UnitedHealth             | UNH    | 0.0008  | 0.0159 | -0.1897 | 0.1204 |
| Visa                     | V      | 0.0008  | 0.0156 | -0.1456 | 0.1397 |
| Verizon                  | VZ     | 0.0003  | 0.0116 | -0.0779 | 0.0887 |
| Walgreens Boots Alliance | WBA    | -0.0002 | 0.0189 | -0.2505 | 0.1187 |
| Walmart                  | WMT    | 0.0005  | 0.0122 | -0.1208 | 0.1107 |

### 3. Data and methodology

#### 3.1. Sample and data

Our empirical analysis focuses on forecasting stock volatility for the constituents of the Dow Jones Industrial Average Index. Our sample includes the daily log returns between October 1, 2012, and June 30, 2024 ( $T = 2,957$  days). We follow [Mattera and Otto \(2024\)](#) to select the recent 12-year data to construct the sample. The stocks with any missing values are removed from the sample, and we are left with 29 nodes/stocks, as shown in [Table 1](#). [Fig. 1a](#) depicts the daily median log returns, while [Fig. 1b](#) shows the median absolute returns of stocks, used as a measure of stock market volatility.

#### 3.2. Sample split and rolling window forecast

Following [Mattera and Otto \(2024\)](#), we employ a rolling window procedure for the forecast. For the log-ARCH model, the k-NN models, and D-vine copula model, we first divide the sample into training and testing sets for each stock. Specifically, the last 500 observations (i.e., the last two years) for out-of-sample testing. The first  $M = 2,457$  observations (the 10-year data before the recent two years) are used as a training dataset to obtain the edge weight matrix  $W$  and to estimate the model parameters. Then, the trained models are used for the one-step-ahead (next-day) forecasts at  $M + 1$ . Then, according to a rolling window procedure, the oldest observation is removed for the next step, and the new realized observation at  $M + 1$  is included in the estimation sample. Parameters except the edge weight matrix  $W$  are re-estimated with the new data, and the forecasts are obtained for  $M + 2$ . This procedure is repeated until no new observation is available and all  $T - M = 500$  volatilities are predicted for each stock.

For the copula entropy model, since it captures the short-term connections, we use the 2-year data before the recent 500 observations as the training dataset to train the edge weight matrix  $W$  and to estimate the initial model parameters. Then the rolling procedure is the same as for the other models.

Forecast accuracy is evaluated using root mean squared forecast error (RMSFE) and mean absolute forecast error (MAFE), with realized squared log returns as a proxy for volatility.

#### 3.3. Forecasting models

We examine a network  $G = (V, E)$  of stock returns, consisting of vertices  $V$  to represent stocks and connections  $E$  between each pair of stocks. Our observations consist of return series  $\{R_{it} : t = 1, \dots, T, r_i \in V\}$ . We aim to predict the volatility of stocks for future points of time  $T + 1, T + 2, \dots$ .

### 3.3.1. Log-ARCH model

We use the univariate log-ARCH model as the baseline to evaluate forecasting models for stock volatility. The *log – ARCH(P)* model (Geweke, 1986) for stock  $i$  is given by:

$$R_{it} = \sqrt{h_{it}}\varepsilon_{it}, i = 1, \dots, n, t = 1, \dots, T, \quad (1)$$

$$\ln h_{it} = \omega_i + \sum_{p=1}^P \gamma_{ip} \ln R_{i(t-p)}^2, \quad (2)$$

where  $h_{it}$  is the conditional variance, and  $\gamma_{ip}$  are the ARCH parameters. Following Otto et al. (2024), we use the lag 1-day information ( $P = 1$ ) to forecast the next-day return. It simplifies model estimation while providing a robust baseline for performance evaluation. However, the method is flexible and can be readily extended to incorporate lag  $P$ -day information.

### 3.3.2. Network based k-NN models

The network log-ARCH model (k-NN models), based on the dynamic spatiotemporal ARCH model (Mattera and Otto, 2024), incorporates dependencies between stocks. The log-volatility of stock  $i$  is influenced by the returns of neighboring stocks, formulated as:

$$h_t^* = \omega + \Gamma R_{t-1}^* + \rho W R_t^*, \quad (3)$$

where  $W$  represents the spatial weight matrix of dependencies,  $\Gamma$  captures stock-specific effects,  $\omega = (\omega_1, \dots, \omega_n)'$  denotes the vector of constant terms and  $\rho$  is a parameter for volatility spillovers.

We also define the connections between stocks (Matrix  $W$ ) to improve volatility estimation. We explore three approaches to measure the distance between stock returns: raw-data-based, feature-based, and model-based (Maharaj et al., 2019). Specifically, Euclidean distance, correlation between time series (Tumminello et al., 2010) and Piccolo distance (Piccolo, 1990) are used.

We compute these three distances for each pair of  $r_{it}$  and  $r_{jt}$ , organizing the results into an  $n \times n$  matrix that is used to construct the spatial weight matrix  $W$ . To improve accuracy, we eliminate unimportant connections using thresholds and entropy-based methods. We also compare our approach to the k-NN method (Otto et al., 2024), which determines the weight matrix as follows:

$$\omega_{ij} = \begin{cases} \frac{1}{k} & \text{if } j \in N_k(i) \\ 0 & \text{otherwise} \end{cases}, \quad (4)$$

where  $N_k(i)$  is the set of  $k$  nearest neighbours of  $r_{it}$ .

### 3.3.3. D-vine copula model

The D-vine copula is a multivariate distribution that models dependencies while maintaining uniform marginals. According to Sklar's theorem (Sklar, 1959), for a random vector  $X = (X_1, \dots, X_d)'$  with the distribution function  $F$ , the associated copula  $C$  satisfies:

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d)), \quad (5)$$

where  $F_1, \dots, F_d$  are marginal distribution functions. The D-vine copula decomposes the joint density into bivariate copulas, structured sequentially in a vine network (Czado, 2010). The D-vine copula-based quantile regression function is:

$$\hat{q}_a(x_1, \dots, x_d) = F_{Y|X_1, \dots, X_d}^{-1}(a|x_1, \dots, x_d), \quad (6)$$

where  $q_a$  estimates the conditional quantile for given inputs. Further details are available in Kraus and Czado (2017).

We employ the D-vine copula approach to identify neighboring stocks based on their marginal distributions. The corresponding weight matrix in this approach is defined as follows:

$$\omega_{ij} = \begin{cases} \frac{1}{|N_{Dvine}(i)|} & \text{if } j \in N_{Dvine}(i) \\ 0 & \text{otherwise} \end{cases}, \quad (7)$$

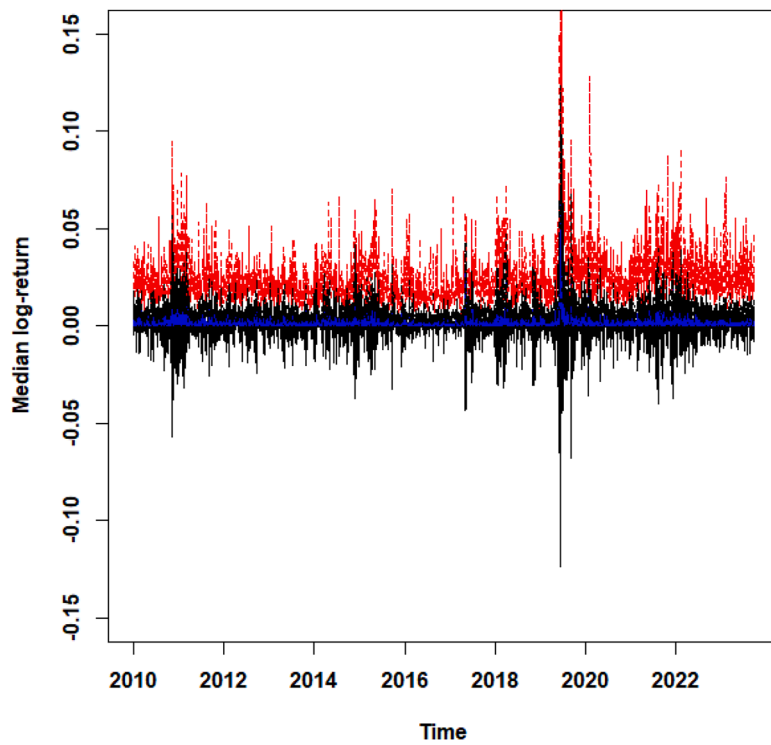
where,  $N_{Dvine}(i)$  represents the set of neighbors of stock  $i$  based on the D-vine copula, and  $|N_{Dvine}(i)|$  represents the cardinality of the  $N_{Dvine}(i)$  set. The matrix  $W$  in Eq. (7) is not symmetric, and each stock's neighbors have the same weight. Instead of defining the  $k$  for the model, the D-vine copula model can automatically remove the redundant neighbors and optimize the number of connected stocks.

### 3.3.4. Copula entropy model

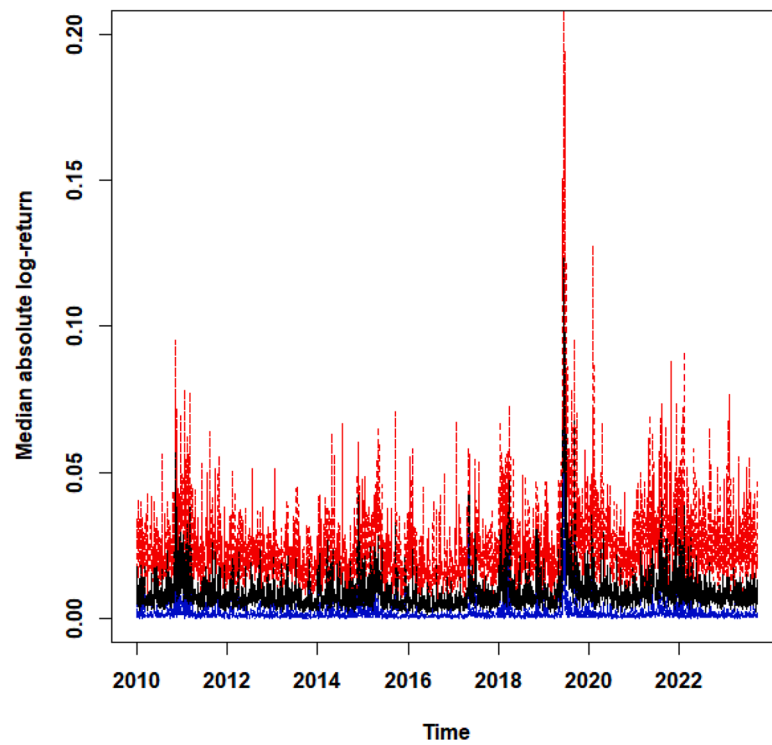
Copula entropy is derived from the copula density function and is closely related to Shannon entropy, as shown in the Eq. (8):

$$H = - \int f(x) \ln f(x) dx = E[\ln f(x)] \quad (8)$$

The feature selection method based on copula-based Mutual Information (MI) (Lall et al., 2021) is formulated as:



(a)



(b)

*(caption on next page)*

**Fig. 1.** Volatility variations based on time series summary of log returns. Top: daily median log returns (solid line) and 5 % and 95 % quantiles (dashed lines). Bottom: daily median absolute log returns (solid line) and 5 % and 95 % quantiles.

$$\begin{aligned} f_{\text{CFS}} &= \arg \max_{f_i \in (F-S)} [I_c(f_i; Y) - I_c(f_i; f_1, \dots, f_s)] \\ &= \arg \max_{f_i \in (F-S)} [-H(C(P(f_i), P(Y))) + H(C(P(f_i), P(f_1), \dots, P(f_s)))], \end{aligned} \quad (9)$$

where  $F$  is the set of non-focus stocks, and  $S$  is the set of selected stocks. The first term measures relevance, while the second term accounts for redundancy.

The method iteratively selects stocks maximizing copula-based MI with the target stock while minimizing redundancy among selected stocks. It identifies an optimal subset of  $k$  neighboring stocks for volatility forecasting, ensuring an effective and stable feature selection process.

### 3.3.5. Hybrid model

The D-vine copula approach captures long-term dependencies using 10-year training data, reflecting stable co-movements and structural relationships. In contrast, the copula entropy method captures short-term dependencies with a 2-year training window, allowing the network to adapt to recent market dynamics. The key distinction lies in the time horizon rather than the dependency type. While the D-vine copula structure remains fixed to preserve long-run information, the copula entropy network evolves over time. Following [Tian and Ji \(2022\)](#), we combine both methods to build a hybrid model that integrates persistent and transient dependencies. The neighbors of each stock in this hybrid model are defined as follows:

$$S_i = S_{i,Dvine} \cup S_{i,Entropy}, \quad (10)$$

where  $S_{i,Dvine}$  and  $S_{i,Entropy}$  denote the sets of neighboring stocks of stock  $i$  extracted from the D-vine copula and copula entropy approaches respectively.

## 4. Results and discussion

### 4.1. Network construction

We construct the matrix  $W$  using three distance measures and copula-based methods, assigning weights based on the k-Nearest Neighbors method (k-NN) and copula methods. [Figs. 2 and 3](#) illustrate the resulting networks, which reveal the stock relationships relevant to volatility forecasting. For example, in the Euclidean network ([Fig. 2a](#)), HON and JNJ influence 18 stocks. BA and CRM, with high volatility, occupy distinct positions. BA affects only 8 stocks, and CRM influences the IT industry and central firms.

### 4.2. Out-of-sample performance

We evaluate the out-of-sample forecasting performance of copula methods in comparison to the log-ARCH model and k-NN models over the out-of-sample period 2022–2024 in [Table 2](#). The results indicate that the k-NNs have higher forecast accuracy than the log-ARCH benchmark model, which is consistent with the findings in [Mattera and Otto \(2024\)](#). Among the k-NN models, the correlation-based network with  $k = 10$  yields the best MAFE, while the Piccolo-based network with  $k = 3$  performs best in terms of RMSFE. However, a key challenge with the k-NN approach is the absence of a universally optimal configuration; no single method consistently excels across both error metrics (RMSFE and MAFE). The D-vine copula model produces lower RMSFE and MAFE relative to the log-ARCH model and most of the k-NN models, which supports the H1 hypothesis.

Among the copula approaches, copula entropy outperforms the D-vine copula when  $k = 10$ . Furthermore, the hybrid model that combines the D-vine copula with copula entropy yields the lowest RMSFE and MAFE, outperforming all individual methods, thereby supporting the H2 hypothesis.

To assess the robustness of the proposed copula-based method, we conduct an additional test during the GFC period when the volatility is more difficult to predict. The out-of-sample period is September 2008 to October 2010, and the model training period is September 1998 to October 2008. We present the results in [Table 3](#). The results show that during this period, the performance of all the models is reduced due to increased uncertainty. However, the D-vine copula models outperform the log-ARCH model and k-NN models, and the hybrid model outperform all the other models. The results show the robust advantages of the copula-based models during different time periods, further supporting H1 and H2 hypotheses.

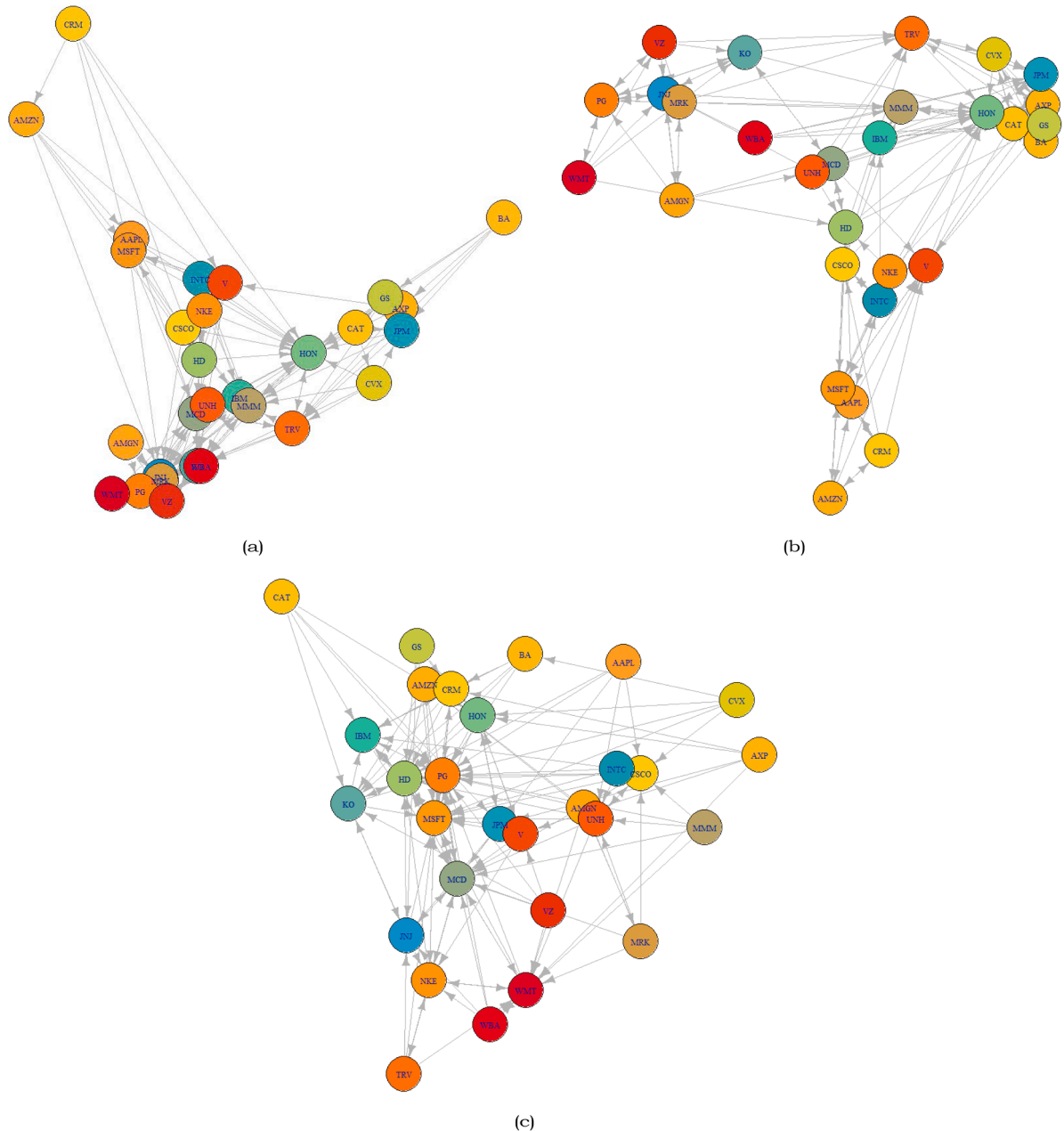
### 4.3. Applications for investment strategy: alpha analysis and volatility groups

To assess the practical implications of the network-based models, we calculate the alpha of individual stocks using the Fama-French five-factor model over a rolling 24-month window. Based on the forecasted volatility, stocks are sorted into five groups. We then construct a long-short portfolio by longing stocks with the lowest forecast volatility and short selling the ones with the highest forecast volatility.

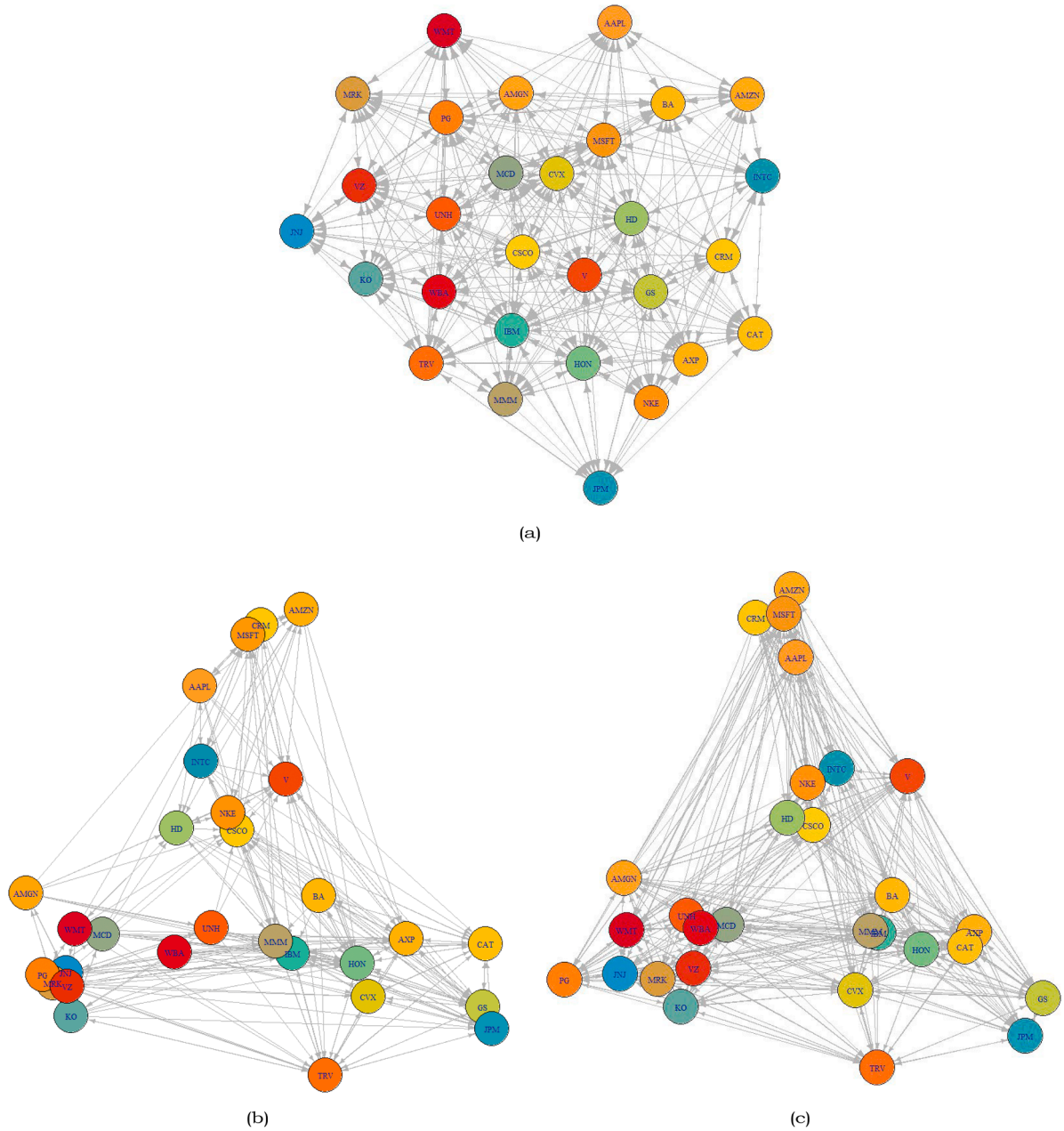
Results in [Table 4](#) show a consistent low-volatility premium, where low-volatility groups outperform high-volatility ones. We use

the Piccolo distance network as the benchmark since it has the highest forecast accuracy among the k-NN models. The alpha difference (L-H) is highest in copula-based networks, particularly the D-vine copula (0.016%) and copula entropy (0.018%), compared to 0.013% in the Piccolo distance network. Copula-based networks are more effective at identifying low-risk, high-return stocks. Moreover, the low-volatility group consistently exhibits return standard deviations that are lower than or comparable to those of the high-volatility group, suggesting reduced downside risk. This effect is particularly pronounced in the hybrid method, which combines the D-vine copula and copula entropy networks.

These findings suggest that integrating alpha analysis with network-based volatility predictions enhances portfolio construction by providing higher alpha and reduced risk.



**Fig. 2.** Networks of stocks constructed using different distance measures and weightings. Nodes are positioned based on their respective distances. The figures illustrate (a) the Euclidean distance network with  $k = 5$ , (b) the correlation distance network with  $k = 5$ , and (c) the Piccolo distance network with  $k = 5$ . Edge colors represent weights, with darker edges indicating higher weights. Arrows denote the direction of influence.



**Fig. 3.** Networks of stocks constructed using different copula methods. Nodes are positioned based on their respective distances. The figures depict (a) the D-vine copula network, (b) the copula entropy network, and (c) the hybrid network. Arrows indicate the direction in which a stock contributes to predicting the volatility of the target stock.

## 5. Conclusion

This study develops a volatility forecasting framework that integrates network structures and copula models within a log-ARCH setting. By leveraging inter-stock connections, the model captures collective dynamics that enhance forecasting accuracy. Among various configurations, copula-based networks outperform traditional distance-based and k-NN models, offering robust performance across different periods and demonstrating strong predictive power without relying on a fixed number of neighbours. The approach also proves valuable in portfolio applications, effectively identifying low-risk, high-return stocks.

Despite its strengths, the model's applicability may be limited in markets with sparse or low-frequency data, and the computational complexity of copula methods could challenge scalability in large datasets. Future research may consider real-time or adaptive network updating, application to intraday or emerging market data, or integration with macroeconomic indicators and sentiment

**Table 2**

Result of forecasting models with different networks from July 2022 to June 2024. The best RMSFE/MAFE in each row is bold. The average stock performance and the p-values of the [Diebold and Mariano \(2002\)](#) and [Clark and West \(2007\)](#) tests are reported. The null hypothesis of both tests is that the benchmark log-ARCH model has better or equal performance than the network log-ARCH with alternative network definitions.

| Model              | log ARCH | k-NN      |        |        |             |        |        |         |        |        | D-vine | Copula entropy |         |        | Hybrid        |
|--------------------|----------|-----------|--------|--------|-------------|--------|--------|---------|--------|--------|--------|----------------|---------|--------|---------------|
|                    |          | Euclidean |        |        | Correlation |        |        | Piccolo |        |        |        |                |         |        |               |
|                    |          | 3         | 5      | 10     | 3           | 5      | 10     | 3       | 5      | 10     |        | 3              | 5       | 10     |               |
| k                  |          | 3         | 5      | 10     | 3           | 5      | 10     | 3       | 5      | 10     |        | 3              | 5       | 10     |               |
| RMSFE (Average)    | 2.8474   | 2.7649    | 2.8195 | 2.5590 | 2.4695      | 2.4284 | 2.4244 | 2.4238  | 2.4278 | 2.4641 | 2.4364 | 2.4847         | 2.4509  | 2.4342 | <b>2.3920</b> |
| P-value of DM test | –        | 0.0000    | 0.0000 | 0.0000 | 0.0000      | 0.0000 | 0.0000 | 0.0000  | 0.0000 | 0.0000 | 0.0000 | 0.0000         | 0.0000  | 0.0000 | 0.0000        |
| P-value of CW test | –        | 0.0000    | 0.0000 | 0.0000 | 0.0000      | 0.0000 | 0.0000 | 0.0000  | 0.0000 | 0.0000 | 0.0000 | 0.0000         | 0.0000  | 0.0000 | 0.0000        |
| MAFE (Average)     | 2.0106   | 2.0760    | 2.0981 | 1.9191 | 1.8645      | 1.8267 | 1.8201 | 1.8205  | 1.8221 | 1.8393 | 1.8199 | 1.8602         | 1.83191 | 1.8137 | <b>1.8097</b> |

**Table 3**  
Result of forecasting models with different networks from September 2008 to August 2010. The best RMSFE/MAFE in each row is bold. The average stock performance and the p-values of the [Diebold and Mariano \(2002\)](#) and [Clark and West \(2007\)](#) tests are reported. The null hypothesis of both tests is that the benchmark log-ARCH model has better or equal performance than the network log-ARCH with alternative network definitions.

| Model              | log ARCH | k-NN      |        |        |             |        |        |         |        |        | D-vine | Copula entropy |        |        | Hybrid        |
|--------------------|----------|-----------|--------|--------|-------------|--------|--------|---------|--------|--------|--------|----------------|--------|--------|---------------|
|                    |          | Euclidean |        |        | Correlation |        |        | Piccolo |        |        |        |                |        |        |               |
|                    |          | 3         | 5      | 10     | 3           | 5      | 10     | 3       | 5      | 10     |        | 3              | 5      | 10     |               |
| k                  |          | 3         | 5      | 10     | 3           | 5      | 10     | 3       | 5      | 10     |        | 3              | 5      | 10     |               |
| RMSFE (Average)    | 4.8953   | 4.8713    | 4.6801 | 3.3909 | 3.8393      | 3.2671 | 3.0657 | 4.1005  | 4.4459 | 4.7960 | 2.8916 | 3.2378         | 3.0168 | 2.9582 | <b>2.8583</b> |
| P-value of DM test | –        | 0.0000    | 0.0000 | 0.0000 | 0.0000      | 0.0000 | 0.0000 | 0.0000  | 0.0000 | 0.0000 | 0.0000 | 0.0000         | 0.0000 | 0.0000 | 0.0000        |
| P-value of CW test | –        | 0.0000    | 0.0000 | 0.0000 | 0.0000      | 0.0000 | 0.0000 | 0.0000  | 0.0000 | 0.0000 | 0.0000 | 0.0000         | 0.0000 | 0.0000 | 0.0000        |
| MAFE (Average)     | 3.1144   | 3.1022    | 3.0828 | 2.4889 | 3.3056      | 3.3025 | 2.1821 | 2.2468  | 2.4822 | 2.4044 | 2.0873 | 4.5804         | 3.5442 | 2.1322 | <b>2.0605</b> |

**Table 4**

Estimated alpha and associated statistics for different stock groups as well as the return spread (L-H) between the lowest and highest groups for D-vine copula, copula entropy, hybrid method and Piccolo network. The table presents the alpha, standard error (S.E.) in parentheses, and the significance levels (*P*-value) (are denoted as \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ ).

| Group    | D-vine              | Entropy              | Hybrid              | Piccolo             |
|----------|---------------------|----------------------|---------------------|---------------------|
| Low (L)  | 0.009***<br>(0.002) | 0.009***<br>(0.002)  | 0.009***<br>(0.002) | 0.009***<br>(0.002) |
| 2        | 0.013***<br>(0.002) | 0.013***<br>(0.002)  | 0.013***<br>(0.002) | 0.015***<br>(0.002) |
| 3        | 0.010***<br>(0.003) | 0.013***<br>(0.003)  | 0.009**<br>(0.003)  | 0.008**<br>(0.003)  |
| 4        | 0.012**<br>(0.004)  | 0.013***<br>(0.004)  | 0.012**<br>(0.004)  | 0.010**<br>(0.004)  |
| High (H) | -0.007**<br>(0.002) | -0.009***<br>(0.002) | -0.004*<br>(0.002)  | -0.003<br>(0.003)   |
| L - H    | 0.016***<br>(0.003) | 0.018***<br>(0.003)  | 0.014***<br>(0.003) | 0.013***<br>(0.004) |

analysis. Combining copula networks with deep learning techniques could further enhance the modelling of complex nonlinear dependencies.

### CRedit authorship contribution statement

**Shahab Nankali:** Writing – original draft, Formal analysis, Data curation, Conceptualization. **Laleh Tafakori:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Mahdi Jalili:** Writing – review & editing, Supervision. **Xiaolu Hu:** Writing – review & editing, Supervision.

### Data availability

Data will be made available on request.

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